

NITHIN KOKKISA

Data Scientist | ML Engineer | AI & Analytics Leader
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EAD Authorization — No Sponsorship Required | Immediately Available

PROFESSIONAL SUMMARY

Data Scientist and AI/ML practitioner with **12 years of applied analytical and digitalization leadership** at HPCL — a Fortune Global 500 energy company (\$50B+ revenue). Built and deployed production ML systems, computer vision pipelines, NLP platforms, RAG-based AI applications, and enterprise analytics platforms across a **55-plant national network** processing **~1.5 million cylinders daily**. Led the data investigation that **saved \$7.7M** — awarded HPCL Best of Best (top 0.25% of 4,000+ officers). Designed **APES** — an AI procurement system reducing evaluation cycles from **2 days to 20 minutes (98.6% reduction)**. Deployed production RAG chatbot on HPCL enterprise portal serving the full national network. IIT Jodhpur B.Tech (Mechanical Engineering) + MBA Finance. Local to DFW. EAD — no sponsorship required.

TECHNICAL SKILLS

Machine Learning & AI: XGBoost, Gradient Boosting, Random Forest, Logistic Regression, K-Means Clustering, Isolation Forest, SHAP Explainability, CLTV Modeling (BG-NBD + Gamma-Gamma), Anomaly Detection

Computer Vision: Python, OpenCV, TensorFlow, CNN-based Defect Detection, OCR Pipelines, Real-time IoT Sensor Analytics

NLP & Generative AI: LangChain, OpenAI GPT-4, FAISS Vector Database, RAG Pipelines, Sentiment Analysis (NLTK, TextBlob, VADER), Text Classification, NER

Forecasting & Statistics: Prophet, SARIMA, ARIMA, ETS, Hierarchical Forecasting, Demand Pattern Classification (ADI/CV²), Forecast Accuracy & Bias (MAPE, WMAPE), A/B Testing, Hypothesis Testing, Regression Analysis, Statistical Process Control

Data Engineering: Python (Pandas, NumPy, Scikit-learn), PySpark ETL, SQL (Window Functions, CTEs, Optimization), SAP-integrated ETL Pipelines, PostgreSQL, SQLite, Git/GitHub, Flask, FastAPI, AWS (EC2, S3, SageMaker), Kubernetes, Docker

Visualization & BI: Power BI (DAX, Star Schema, Time Intelligence), Plotly, Matplotlib, Seaborn, Streamlit, Role-based Dashboards

ERP & Enterprise Systems: SAP S/4HANA (MM/PM) — 2 years, JD Edwards ERP (MM/PM/SD) — 10 years, SCADA/PLC, Automated Tank Gauging, Master Data Management

PROFESSIONAL EXPERIENCE

Data Scientist & Analytics Lead

Hindustan Petroleum Corporation Limited (HPCL)

India Jul 2013 – May 2025

Built and deployed production ML systems, computer vision pipelines, NLP platforms, RAG-based AI applications, and enterprise analytics across a 55-plant national LPG network. Led the \$7.7M data investigation awarded **HPCL Best of Best — top 0.25% of 4,000+ officers**. Managed **435+ personnel and \$3.9M annual budget**. Youngest appointee to senior plant leadership in HPCL history.

AI & Generative AI:

- Designed and deployed **APES (Automated Procurement Evaluation System)** using **LangChain and GPT-4** — automating end-to-end tender evaluation with three-tier human-in-the-loop workflow and complete audit trail. Deployed on ****AWS (EC2, S3)**** and orchestrated with ****Kubernetes.**** Reduced evaluation cycle from **2 days to 20 minutes — 98.6% reduction**.

- Deployed **production RAG-based chatbot** on MyHPCL enterprise portal — enabling natural language querying across Operations, Maintenance, Procurement, and Safety manuals using **LangChain, GPT-4, FAISS vector database**. Built conversation caching layer and human feedback loop for continuous model improvement. Deployed on **AWS (EC2, SageMaker)** with **Kubernetes** orchestration — network-wide across all 55 plants.

Computer Vision & IoT:

- Designed and deployed **OCR-based tare weight automation pipeline (Python, OpenCV, TensorFlow)** achieving **99%+ read accuracy** at 3,200 cylinders/hour across **~1.5 million daily cylinder operations** — eliminating manual encoding errors and systematic overfill/underfill incidents.
- Deployed **CNN-based automated cylinder pre-inspection system** replacing manual visual inspection at 3,200 cylinders/hour — trained on labeled defect datasets (dents, cuts, corrosion) enforcing consistent standards-compliant rejection at full industrial throughput across 55 plants.
- Implemented **IoT sensor-based bottom corrosion detection** using **Isolation Forest anomaly detection (scikit-learn)** — closing critical safety blind spot completely undetectable by human inspection across **~1.5 million daily operations**.

Machine Learning & Predictive Analytics:

- Led **\$7.7M multi-source data investigation** — analyzing 10 years of operational records applying **multivariate correlation analysis** to disprove upstream supplier's \$7.7M financial claim. Demonstrated terminal gain/loss was a function of all operational variables combined — not a single source. **Saved \$7.7M. Awarded Best of Best.**
- Deployed **network-wide predictive maintenance system** across 55 plants — replacing manual gauge monitoring with **continuous IoT sensor data streams** processed through **Isolation Forest anomaly detection (scikit-learn)** with dual-layer rule-based and ML alerting. Reduced unplanned equipment downtime network-wide.
- Built **distributor churn risk prediction model (XGBoost, SHAP)** across **6,200+ HP Gas distributors** — flagging at-risk distributors based on digital feedback trend deterioration across **85 million+ customer touchpoints**.
- Applied **CLTV modeling (BG-NBD + Gamma-Gamma)** and **K-Means segmentation** to stratify 6,200+ distributors by long-term revenue contribution — informing discount tier eligibility and targeted engagement strategy.
- Conducted **network-wide purchase price variance analysis** across 30-40 spare categories — **Python, Pandas, statistical analysis** on multi-year SAP procurement data. Designed and implemented **centralized procurement model** delivering **~\$2.3M in verified annual network-wide savings — 2.6% cost reduction per plant**.

NLP & Sentiment Analytics:

- Built **NLP-based distributor performance analytics platform** — applying **sentiment analysis (NLTK, TextBlob, VADER)** to classify digital customer feedback across delivery, staff behavior, showroom service, and complaint resolution dimensions across **6,200+ distributors**. Integrated sentiment scores into automated **100-point performance scoring model** replacing manual officer evaluation.
- Conducted **safety audit analytics** across 13 plants — ingesting 334 unstructured audit observations, performing **feature engineering and feature construction (Pandas, NumPy)**, pattern analysis via **Matplotlib/Seaborn**, delivering prioritized compliance action framework via **Power BI dashboards**. Achieved **58% improvement in audit compliance** against company benchmark.

Forecasting & Supply Chain Analytics:

- Led end-to-end design of **daily demand forecasting and supply optimization pipeline** across 55 plants — building **Python-based time-series models (Prophet, SARIMA)** with fully **automated SAP-integrated ETL pipeline** triggered on plant day-end completion, delivering next-day indent recommendations to plant, regional, zone, and HQ by **1100 hrs daily**. Achieved **zero supply disruptions across 55-plant network over 12 years** serving **3,300+ distributors**.

- Applied **distributor discount optimization** using **regression modeling, A/B testing, Market Basket Analysis, and CLTV stratification** across 6,200+ distributors — replacing manual officer-driven discount calculations with standardized, data-driven pricing engine eliminating inconsistencies across the network.

Data Engineering & Dashboards:

- Designed and deployed **real-time web-based carousel analytics platform** — **Python (Flask), PostgreSQL backend**, per-filling-head deviation analysis using **Pandas and Plotly** with role-based dashboards from plant officer to HQ replacing end-of-shift manual reports.
- Built **network-wide preventive maintenance platform** — digitizing **~3,000 equipment tags** across 55 plants with OEM schedules, automated due-date alerting, and **role-based Power BI compliance dashboards** from plant to HQ tracking completed vs missed tasks.
- Built **network-wide spare parts warehouse management system** — managing **1,500+ SKUs** with optimum inventory models, **ABC/VED/JIT classification (Python, SQL)**, cross-plant spare transfer engine, and automated purchase initiation alerts.
- Led design and deployment of **automated logistics optimization system** across all plants — replacing manual truck-to-delivery assignment with rule-based automated allocation eliminating planning bias and improving transporter management consistency network-wide.
- Conceptualized and deployed **kiosk-based digital training management system** — automated platform with integrated pre/post **assessment analytics, competency tracking, and training need identification**. Eliminated **6 employee-hours per plant per month**. Delivered compliance dashboards and audit-ready training records to plant management.

GITHUB PORTFOLIO

github.com/Kokkisa — Applied data science portfolio covering ML, NLP, time-series forecasting, PySpark, generative AI, computer vision, and analytics engineering — all rooted in Fortune Global 500 operational problem sets.

EDUCATION

B.Tech, Mechanical Engineering — Indian Institute of Technology (IIT) Jodhpur (2009 – 2013)

MBA, Finance — Osmania University, Hyderabad

ADDITIONAL INFORMATION

Work Authorization: EAD — authorized to work in the U.S., no sponsorship required. Immediately available.

Location: Lewisville, TX (DFW) — open to relocation across the United States.

Languages: English (Professional), Hindi (Native), Telugu (Native), Kannada (Conversational)

Key Recognition: HPCL Best of Best — Outstanding Achievement Award (2019-20) — Top 10 of 4,000+ officers (top 0.25%)